

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

ELECTRIC PARK BRAKE CONTROL MODULE (EPBCM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Electric Park Brake Control Module (EPBCM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Parking Brake](#) (206-05 Parking Brake

and Actuation, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1111-77	Electronic Park Brake Enable - Commanded position not reachable	NOTE: Fault is set when at least one brake is in brake state "unknown" or one brake is applied or fully applied while the other brake is released or fully released - Electric park brake left or right actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Electric park brake left or right actuator internal failure - Rear left or rear right brake caliper internal failure	- Refer to the electrical circuit diagrams and check the electric park brake left and right actuator circuits for short circuit to ground, short circuit to power, open circuit, high resistance - Test the operation of the left and right electric park brake actuators - Test the operation of the rear left and rear right brake calipers
C1059-62	Left And Right Actuator - Internal Temperature Sensors - Signal compare failure	NOTE: Circuit reference TEMP LH / TEMP RH - Electric park brake control module connector pins corroded - Electric park brake left motor connector	

		pins corroded ▪ Electric park brake left actuator temperature sensor circuit, high resistance ▪ Electric park brake right motor connector pins corroded ▪ Electric park brake right actuator temperature sensor circuit, high resistance	NOTE: For this DTC to set the vehicle must be driven for a period of 10 minutes at 10kph. If the vehicle speed falls below 10kph the setting period must be done again. If after the setting period the temperature between the left and right actuators exceeds 30deg°C and one of the brakes is fully applied, the DTC will set ▪ Refer to the electrical circuit diagrams and check the electric park brake control module connector pins for corrosion ▪ Refer to the electrical circuit diagrams and check the electric park brake left motor connector pins for corrosion ▪ Refer to the electrical circuit diagrams and check the electric park brake left actuator temperature sensor circuit for short circuit to ground, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the electric park brake right motor connector pins for corrosion ▪ Refer to the electrical circuit diagrams and check the electric park brake right actuator temperature sensor circuit for short circuit to ground, open circuit, high resistance
C2005-09	Right Actuator - Component failures	▪ Electric park brake right actuator incorrectly installed ▪ Electric park brake right actuator internal failure	▪ Check that the electric park brake right actuator is correctly installed ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the

			fault persists, install a new electric park brake right actuator as required
C2005-71	Right Actuator - Actuator stuck	▪ Electric park brake right actuator incorrectly installed ▪ Electric park brake right actuator internal failure	▪ Check that the electric park brake right actuator is correctly installed ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake right actuator as required
C2005-77	Right Actuator - Commanded position not reachable	▪ Low power supply voltage, battery/charging system fault ▪ Electric park brake right actuator incorrectly installed ▪ Electric park brake right actuator circuit open circuit, high resistance ▪ Electric park brake control module ground circuit open circuit, high resistance ▪ Electric park brake right actuator internal failure	NOTE: This DTC will set after detecting disc contact, but full clamping force is not reached within three seconds of the brake being applied. Or, if the target disc clamping force is not achieved within 25 seconds from start of park brake application ▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Check that the electric park brake right actuator is correctly installed. If the actuator is correctly installed, release and apply the electric parking brake several times ▪ Refer to the electrical circuit diagrams and check the electric park brake right actuator circuit for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check datalogger signal - Right Park Brake Motor Current (0x2B34) - and operate the electric park brake switch. If the

			signal is <15A refer to the electrical circuit diagrams and check the electric park brake control module ground circuits for open circuit, high resistance ▪ If the fault persists, install a new electric park brake right actuator as required
C2005-86	Right Actuator - Signal invalid	▪ Electric park brake right actuator temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric park brake right actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric park brake right actuator temperature sensor internal failure	▪ Refer to the electrical circuit diagrams and check the electric park brake right actuator temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the electric park brake right actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake right actuator temperature sensor as required
C2005-95	Right Actuator - Incorrect assembly	▪ Electric park brake right actuator incorrectly installed ▪ Rear right brake disc or pads incorrectly installed ▪ Electric park brake right actuator internal failure	▪ Check that the electric park brake right actuator is correctly installed ▪ Check that the rear right brake disc and pads are correctly installed ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake right actuator as required
C2006-09	Left Actuator - Component failures	▪ Electric park brake left actuator incorrectly installed ▪ Electric park brake	▪ Check that the electric park brake left actuator is correctly installed ▪ Using the manufacturer

		left actuator internal failure	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake left actuator as required
C2006-71	Left Actuator - Actuator stuck	▪ Electric park brake left actuator incorrectly installed ▪ Electric park brake left actuator internal failure	▪ Check that the electric park brake left actuator is correctly installed ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake left actuator as required
C2006-77	Left Actuator - Commanded position not reachable	▪ Low power supply voltage, battery/charging system fault ▪ Electric park brake left actuator incorrectly installed ▪ Electric park brake left actuator circuit open circuit, high resistance ▪ Electric park brake control module ground circuit open circuit, high resistance ▪ Electric park brake left actuator internal failure	NOTE: This DTC will set after detecting disc contact, but full clamping force is not reached within three seconds of the brake being applied. Or, if the target disc clamping force is not achieved within 25 seconds from start of park brake application ▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Check that the electric park brake left actuator is correctly installed. If the actuator is correctly installed, release and apply the electric parking brake several times ▪ Refer to the electrical circuit diagrams and check the electric park brake left actuator circuit for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check datalogger signal - Left Park Brake Motor Current

			(0x2B33) - and operate the electric park brake switch. If the signal is <15A refer to the electrical circuit diagrams and check the electric park brake control module ground circuits for open circuit, high resistance ▪ If the fault persists, install a new electric park brake left actuator as required
C2006-86	Left Actuator - Signal invalid	▪ Electric park brake left actuator temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric park brake left actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric park brake left actuator temperature sensor internal failure	▪ Refer to the electrical circuit diagrams and check the electric park brake left actuator temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the electric park brake left actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake left actuator temperature sensor as required
C2006-95	Left Actuator - Incorrect assembly	▪ Electric park brake left actuator incorrectly installed ▪ Rear left brake disc or pads incorrectly installed ▪ Electric park brake left actuator internal failure	▪ Check that the electric park brake left actuator is correctly installed ▪ Check that the rear left brake disc and pads are correctly installed ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake left actuator as required
C2007-01	Right Motor - General electrical failure		▪ Refer to the electrical circuit diagrams and check the electric parking brake control module, right power supply fuse and

		NOTE: Circuit reference - RH MOTOR + / RH MOTOR - / VBATT MOTOR RH / RH MOTOR GND ▪ Fuse failure ▪ Electric park brake right actuator circuit open circuit, high resistance ▪ Electric park brake control module right actuator power or ground circuit open circuit, high resistance ▪ Electric park brake control module connectors damaged or corroded ▪ Electric park brake right actuator internal failure ▪ Electric park brake control module internal failure	circuit ▪ Refer to the electrical circuit diagrams and check the electric park brake right actuator circuit for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the electric park brake control module right actuator power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the electric park brake control module connectors for damage or corrosion ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake right actuator as required ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module as required
C2007-11	Right Motor - Circuit short to ground	NOTE: Circuit reference RH MOTOR +VE / RH MOTOR - VE ▪ Electric park brake right actuator circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the electric park brake right actuator circuit for short circuit to ground ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake right actuator as required ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the

		▪ Electric park brake right actuator internal failure ▪ Electric park brake control module internal failure	fault persists, install a new electric park brake control module as required
C2007-12	Right Motor - Circuit short to battery	NOTE: Circuit reference - RH MOTOR + / RH MOTOR - ▪ Electric park brake right actuator circuit short circuit to power ▪ Electric park brake control module connector short circuit to power ▪ Electric park brake control module connector pins corroded ▪ Electric park brake control module internal failure	▪ Refer to the electrical circuit diagrams and check the electric park brake right actuator circuit for short circuit to power ▪ Refer to the electric circuit diagrams and check the electric park brake control module connector for short circuit to power ▪ Refer to the electrical circuit diagrams and check the electric parking brake control module connector pins for corrosion ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module as required
C2007-13	Right Motor - Circuit open	NOTE: Circuit reference - VBATT MOTOR RH ▪ Fuse failure ▪ Electric park brake control module right actuator power circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the electric park brake control module right power supply fuse and circuit ▪ Refer to the electrical circuit diagrams and check the electric park brake control module right actuator power circuit for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the electric park brake control module connectors for damage or corrosion ▪ Using the manufacturer

		▪ Electric park brake control module connectors damaged or corroded ▪ Electric park brake control module internal failure	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module as required
C2007-16	Right Motor - Circuit voltage below threshold	NOTE: Circuit reference - RH MOTOR + / RH MOTOR - / VBATT MOTOR RH / RH MOTOR GND ▪ Fuse failure ▪ Electric park brake right actuator circuit open circuit, high resistance ▪ Electric park brake control module connectors damaged or corroded ▪ Electric park brake control module right actuator power or ground circuit open circuit, high resistance ▪ Electric park brake right actuator internal failure ▪ Battery/charging system fault ▪ Park brake control module internal failure	▪ Refer to the electrical circuit diagrams and check the electric park brake control module right power supply fuse and circuit ▪ Refer to the electrical circuit diagrams and check the electric park brake right actuator circuit for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the electric park brake control module connectors for damage or corrosion ▪ Refer to the electrical circuit diagrams and check the electric park brake control module right actuator power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake right actuator as required ▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module as required
C2007-	Right Motor -		▪ Refer to the electrical circuit

18	Circuit current below threshold	NOTE: Circuit reference RH MOTOR +VE / RH MOTOR -VE / VBATT MOTOR RH / GND MOTOR R - Electric park brake right actuator circuit open circuit, high resistance - Electric park brake control module right actuator power or ground circuit open circuit, high resistance - Electric park brake right actuator internal failure - Battery/charging system fault	diagrams and check the electric park brake right actuator circuit for open circuit, high resistance - Refer to the electrical circuit diagrams and check the electric park brake control module right actuator power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake right actuator as required - Refer to the relevant section of the workshop manual and test the battery and charging system
C2007-19	Right Motor - Circuit current above threshold	NOTE: Circuit reference RH MOTOR +VE / RH MOTOR -VE - Electric park brake right actuator circuit short circuit to ground, short circuit to power - Electric park brake right actuator internal failure	- Refer to the electrical circuit diagrams and check the electric park brake right actuator circuit for short circuit to ground, short circuit to power - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake right actuator as required

C2007-1D	Right Motor - Circuit current out of range	▪ Electric park brake right actuator internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake right actuator as required
C2008-01	Left Motor - General electrical failure	NOTE: Circuit reference LH MOTOR +VE / LH MOTOR -VE / VBATT MOTOR LH / GND MOTOR L ▪ Electric park brake left actuator circuit open circuit, high resistance ▪ Electric park brake control module left actuator power or ground circuit open circuit, high resistance ▪ Electric park brake control module connector damaged or corroded ▪ Electric park brake left actuator internal failure ▪ Electric park brake control module internal failure	▪ Refer to the electrical circuit diagrams and check the electric park brake left actuator circuit for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the electric park brake control module left actuator power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the electric park control module connector for damage or corrosion ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake left actuator as required ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module as required
C2008-11	Left Motor - Circuit short to ground		▪ Refer to the electrical circuit diagrams and check the electric park brake left actuator circuit for short circuit to ground ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the

		NOTE: Circuit reference LH MOTOR +VE / LH MOTOR -VE ▪ Electric park brake left actuator circuit short circuit to ground ▪ Electric park brake left actuator internal failure ▪ Electric park brake control module internal failure	fault persists, install a new electric park brake left actuator as required ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module as required
C2008-12	Left Motor - Circuit short to battery	NOTE: Circuit reference LH MOTOR +VE / LH MOTOR -VE ▪ Electric park brake left actuator circuit short circuit to power ▪ Electric park brake control module connector short circuit to power ▪ Electric park brake module internal failure	▪ Refer to the electrical circuit diagrams and check the electric park brake left actuator circuit for short circuit to power ▪ Refer to the electrical circuit diagrams and check the electric park brake control module connector for short circuit to power ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module as required
C2008-13	Left Motor - Circuit open		▪ Refer to the electrical circuit diagrams and check the park brake control module left power supply fuse and circuit ▪ Refer to the electrical circuit diagrams and check the electric park brake control module left

		NOTE: Circuit reference - VBATT MOTOR LH ▪ Fuse failure ▪ Electric park brake control module left actuator power circuit open circuit, high resistance ▪ Electric park brake control module connector damaged or corroded ▪ Electric park brake module internal failure	actuator power circuit for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the electric park brake control module connector for damage or corrosion ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module as required
C2008-16	Left Motor - Circuit voltage below threshold	NOTE: Circuit reference - LH MOTOR + / LH MOTOR - / VBATT MOTOR LH / LH MOTOR GND ▪ Fuse failure ▪ Electric park brake left actuator circuit open circuit, high resistance ▪ Electric park brake control module left actuator power or ground circuit open circuit, high resistance ▪ Electric park brake control module	▪ Refer to the electrical circuit diagrams and check the park brake control module left power supply fuse and circuit ▪ Refer to the electrical circuit diagrams and check the electric park brake left actuator circuit for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the electric park brake control module left actuator power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the electric park brake control module connector for damage or corrosion ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake left actuator as required

		connector damaged or corroded ▪ Electric park brake left actuator internal failure ▪ Battery/charging system fault ▪ Electric park brake control module internal failure	▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module as required
C2008-18	Left Motor - Circuit current below threshold	NOTE: Circuit reference LH MOTOR +VE / LH MOTOR -VE / VBATT MOTOR LH / GND MOTOR L ▪ Electric park brake left actuator circuit open circuit, high resistance ▪ Electric park brake control module left actuator power or ground circuit open circuit, high resistance ▪ Electric park brake left actuator internal failure ▪ Battery/charging system fault	▪ Refer to the electrical circuit diagrams and check the electric park brake left actuator circuit for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the electric park brake control module left actuator power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake left actuator as required ▪ Refer to the relevant section of the workshop manual and test the battery and charging system
C2008-19	Left Motor - Circuit current above threshold		▪ Refer to the electrical circuit diagrams and check the electric park brake left actuator circuit for short circuit to ground, short circuit to power or wires shorted together ▪ Refer to the electrical circuit

		NOTE: Circuit reference - LH MOTOR + / LH MOTOR - ▪ Electric park brake left actuator circuit short circuit to ground, short circuit to power or wires shorted together ▪ Electric park brake control module connector short circuit to ground, short circuit to power ▪ Electric park brake left actuator internal failure	diagrams and check the electric park brake control module connector for short circuit to ground, short circuit to power ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake left actuator as required
C2008-1D	Left Motor - Circuit current out of range	▪ Electric park brake left actuator internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake left actuator as required
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	▪ Invalid data received from another control module via the high speed CAN bus (powertrain)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-82	High Speed CAN Communication Bus - Alive/sequence counter incorrect / not updated	▪ Invalid data received from another control module via the high speed CAN bus (powertrain)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index

U0001-87	High Speed CAN Communication Bus - Missing message	Missing message from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-88	High Speed CAN Communication Bus - Bus off	High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-00	Internal Control Module Software Incompatibility - No sub type information	- Car configuration file mismatch with vehicle specification - Incorrect electric park brake control module installed	- Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary - Install a new electric park brake control module as necessary
U1A4C-00	Build / End of Line Mode Active - No sub type information	NOTE: This DTC is for use during production only Electric park brake control module in build / end of line mode	Using the manufacturer approved diagnostic system, clear the DTCs and retest
U1A4C-68	Build / End of Line Mode Active - Event information		Using the manufacturer approved diagnostic system, clear the DTCs and retest

		NOTE: This DTC is for use during production only ▪ Electric park brake control module in build / end of line mode	
U2002-13	Switch - Circuit open	NOTE: Circuit reference - SW1 / SW2 / SW3 / SW4 ▪ Electric park brake switch poor connection or disconnected ▪ Electric park brake switch circuit open circuit, high resistance ▪ Electric park brake switch internal failure	▪ Refer to the electrical circuit diagrams and check the electric park brake switch connector for poor connection or being disconnected ▪ Refer to the electrical circuit diagrams and check the electric park brake switch circuit for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear DTCs and retest. If the fault persists, install a new electric park brake switch as required
U2002-64	Switch - Signal plausibility failure	NOTE: Circuit reference SW1 / SW2 / SW3 / SW4 ▪ Electric park brake switch circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the electric park brake switch circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Test the operation of the electric park brake switch ▪ Using the manufacture approved diagnostic system, clear DTCs and retest. If the fault persists, install a new electric park brake switch as required ▪ Using the manufacture

		▪ Electric park brake switch stuck active ▪ Electric park brake switch internal failure ▪ Electric park brake control module internal failure	approved diagnostic system, clear DTCs and retest. If the fault persists, install a new electric park brake control module as required
U2002-67	Switch - Signal incorrect after event	▪ Electric park brake switch stuck active ▪ Electric park brake switch internal failure	▪ Test the operation of the electric park brake switch ▪ Install a new electric park brake switch as necessary
U2002-92	Switch - Performance or incorrect operation	NOTE: Circuit reference - SW1 / SW2 / SW3 / SW4 ▪ Electric park brake switch needs to reset ▪ Electric park brake switch contaminated ▪ Electric park brake switch connector damaged, corroded or backed out terminals ▪ Electric park brake switch circuits short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric park brake switch internal failure	NOTE: Prior to carrying out any repair procedures, operate the electric park brake switch to the brake apply, brake release and neutral positions. If the fault does not clear carry out the following repair actions below ▪ With the ignition set to on, apply the electric park brake for a minimum of 10 seconds, then release. Clear the DTC and retest ▪ Check for any contamination/debris that may be causing the electric park brake switch to be stuck ▪ Disconnect the electric park brake switch and check for connector damage, corroded or backed out terminals. Clear the DTC and retest ▪ Refer to the electrical circuit diagrams and check the electric park brake switch circuits for short circuit to ground, short circuit to power, open circuit, high resistance

			Using the manufacturer approved diagnostic system, clear DTCs and retest. If the fault persists, install a new electric park brake switch as required
U2012-00	Car Configuration Parameter(s) - No sub type information	- Car configuration file mismatch with vehicle specification - High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Incorrect electric park brake control module installed	- Using the manufacturer approved diagnostic system, check and up-date the car configuration file as required - Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system clear DTCs and retest. If the fault persists, install a new electric park brake control module as required
U2016-05	Control Module Main Software - System programming failures	Electric park brake control module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module as required
U2100-54	Initial Configuration Not Complete - Missing calibration	NOTE: This DTC is for use during production only Electric park brake control module assembly check not completed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module as required

U3000-00	Control Module - No sub type information	▪ Electric park brake control module incorrectly installed ▪ Loose mounting may cause the internal accelerometer to output an implausible signal and set this DTC ▪ Electric park brake control module internal failure	▪ Install the electric park brake control module correctly ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module as required
U3000-13	Control Module - Circuit open	▪ Electric park brake control module power or ground circuit open circuit, high resistance ▪ Battery/charging system fault	▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and check the electric park brake control module power and ground circuits for open circuit, high resistance ▪ Refer to the relevant section of the workshop manual and test the battery and charging system
U3000-29	Control Module - Signal invalid	▪ Electric park brake control module incorrectly installed ▪ Loose mounting may cause the internal accelerometer to output an implausible signal and set this DTC ▪ Electric park brake control module internal failure	▪ Install the electric park brake control module correctly ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module
U3000-48	Control Module - Supervision software failure	▪ Electric park brake control module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control

			module as required
U3000-54	Control Module - Missing calibration	▪ Electric park brake control module is not configured correctly	NOTE: Ensure the vehicle is parked on level ground and the electric park brake control module mounting position is correct, before trying to calibrate ▪ Using the manufacturer approved diagnostic system, re-configure the electric park brake control module with the latest level software
U3000-55	Control Module - Not configured	▪ Electric park brake control module is not configured correctly	▪ Using the manufacturer approved diagnostic system, re-configure the electric park brake control module with the latest level software
U3000-78	Control Module - Alignment or adjustment incorrect	▪ Electric park brake control module mounting position incorrect ▪ Electric park brake control module longitudinal accelerometer calibration incorrect ▪ Electric park brake control module internal failure	▪ Check that the Electric park brake control module is mounted correctly. Rectify if needed ▪ Ensure that vehicle is parking on level surface and free from vibration. Run the calibration routine Parking Brake - Longitudinal accelerometer calibration (0x211D) from the manufacturer approved diagnostic tool. Clear DTC and retest ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric park brake control module as required
U3006-16	Control Module Input Power A -	▪ Electric park brake control module power or ground	▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU

	Circuit voltage below threshold	circuit open circuit, high resistance ▪ Battery/charging system fault	Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and check the electric park brake control module power and ground circuits for open circuit, high resistance ▪ Refer to the relevant section of the workshop manual and test the battery and charging system
U3006-17	Control Module Input Power A - Circuit voltage above threshold	▪ Battery/charging system fault	▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the relevant section of the workshop manual and test the battery and charging system
U300A-62	Ignition Switch - Signal compare failure	NOTE: Circuit reference IGN ▪ Mismatch between the hardwired ignition signal to the electric park brake control module and the value broadcast on the powertrain high speed CAN bus	▪ Refer to the electrical circuit diagrams and check the electric park brake control module ignition signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance